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Economics Fundamentals
Level: Beginner to Intermediate • Rev. 2026

Supply, Demand, and Price Elasticity in Real Markets

Summary: How buyer willingness to pay interacts with producer cost to determine equilibrium pricing, market clearing, and revenue elasticity.

Learning Objectives

- Explain the Law of Demand and the Law of Supply.
- Identify market clearing equilibrium and diagnose shortages versus surpluses.
- Calculate Price Elasticity of Demand (PED) using the midpoint formula.
- Use elasticity to predict whether price increases will raise or lower total revenue.

Core Principles & Key Takeaways

- Market equilibrium is where quantity demanded equals quantity supplied.
- Elastic products ($|\text{PED}| > 1$) lose total revenue when prices rise.
- Inelastic products ($|\text{PED}| < 1$) gain total revenue when prices rise.
- Pricing decisions must evaluate customer substitutes, switching costs, and necessity.

1. Mechanics of Supply, Demand, and Equilibrium

In competitive markets, price serves as an informational signal coordinating consumption and production:

* **Law of Demand:** Ceteris paribus (all else held equal), as the unit price of a good rises, consumer quantity demanded falls, and vice versa.
* **Law of Supply:** As market price rises, suppliers are willing to produce and sell more units to capture higher marginal profits.
* **Market Equilibrium:** The price point (P^*) where quantity demanded equals quantity supplied ($Q_d = Q_s$). At this price, the market clears without excess inventory (surplus) or unfulfilled buyer queues (shortage).

2. Price Elasticity of Demand (PED)

Price Elasticity of Demand (PED) measures the responsiveness of customer demand to changes in price:

$$\text{PED} = \frac{\% \Delta Q_d}{\% \Delta P} = \frac{(Q_2 - Q_1) / \left(\frac{Q_1 + Q_2}{2}\right)}{(P_2 - P_1) / \left(\frac{P_1 + P_2}{2}\right)}$$

Interpreting Elasticity Values:
1. **$|\text{PED}| > 1$ (Elastic Demand):** Consumers are highly price-sensitive. A 10% price hike causes sales volume to drop by more than 10%. **Raising prices will decrease Total Revenue.**
2. **$|\text{PED}| < 1$ (Inelastic Demand):** Consumers have few substitutes or the product is an essential necessity. A 10% price hike causes volume to fall by less than 10%. **Raising prices will increase Total Revenue.**

3. **$|\text{PED}| = 1$ (Unitary Elasticity):** Percentage changes in price and volume are identical; Total Revenue remains constant.

3. African Market Case Study: Food Staples vs. Luxury Services

Consider consumer behavior in urban Nigeria:

* **Inelastic Good:** Petroleum fuel or fundamental staple grains (rice/garri). Because substitutes are limited and everyday survival relies on energy/food, when prices rose sharply post-subsidy removal in 2023, consumption dropped only modestly ($|\text{PED}| \approx 0.35$). Total merchant revenue rose despite lower physical volume.
* **Elastic Good:** Fine dining restaurants or designer shoes. When inflation compressed consumer disposable income, discretionary spending on fine dining dropped drastically ($|\text{PED}| \approx 2.4$), prompting restaurateurs to offer discounted lunch bowls to preserve customer volume.

4. Step-by-Step Elasticity Calculation

Problem: An e-commerce platform in Lagos sells a wireless phone accessory.
* Original: Price $P_1 = ₦10,000$, Units sold $Q_1 = 1,200$ monthly. Total Revenue = $1,200 \times ₦10,000 = ₦12,000,000$.
* New: Price increases to $P_2 = ₦12,000$, Units sold fall to $Q_2 = 900$ monthly. Total Revenue = $900 \times ₦12,000 = ₦10,800,000$.

Step 1: Calculate percentage changes using the midpoint formula:
$$\% \Delta Q = \frac{900 - 1200}{(900 + 1200) / 2} = \frac{-300}{1050} = -28.57\%$$
$$\% \Delta P = \frac{12000 - 10000}{(12000 + 10000) / 2} = \frac{2000}{11000} = +18.18\%$$

Step 2: Calculate PED:
$$\text{PED} = \frac{-28.57\%}{+18.18\%} = -1.57$$

Interpretation:
Because $|\text{PED}| = 1.57 > 1$, the product is **price-elastic**. The price increase caused volume to drop significantly, cutting monthly revenue from ₦12.0M to ₦10.8M. The company should reconsider this price hike or differentiate the product to reduce price sensitivity.

Self-Assessment & Review Questions

Q1: If a software firm increases subscription rates by 15% and user cancellations cause subscriber count to fall by only 3%, what is the demand elasticity?

A. Elastic (|PED| = 5.0)
B. Inelastic (|PED| = 0.20)
C. Unitary (|PED| = 1.0)
D. Perfectly elastic

Correct Answer: B • |PED| = 3% / 15% = 0.20. Since it is below 1, demand is inelastic, and total recurring revenue will expand following the price adjustment.

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